

Health, Longevity & Fitness: The State of the Field

*An unbiased, evidence-graded synthesis tied to the biophysics canon —
where the science is strong, where it is sold harder than it is true.*

Prepared by Nucleus Brain (AI orchestrator) · AG Farms Venture Studio

Corpus [_intake/health-longevity-fitness](#) · bead [bkt-bg6](#)

Method 5 waves · ~15 parallel research agents · OpenAlex / PubMed / Europe PMC / ClinicalTrials.gov / YouTube

Date 28 June 2026 · **Doctrine** index-all · grade-everything · mechanism ≠ outcome ≠ protocol

247	176	35	14	24
GRADED CLAIMS	PEOPLE CARDED	LIVE CONFLICTS	DOMAINS	LABS MAPPED

Abstract & reading instructions

This report distills a 247-claim research corpus on human healthspan into a single intellectually honest map. Its central finding is structural, not a discovery: **the field's evidence is lopsidedly distributed against its attention**. The interventions with the largest, most replicated effects are behavioral and dull (don't smoke; build fitness and strength; move; sleep; keep apoB low; stay connected). The interventions that command the market and the podcast hours (NAD⁺, senolytics, cold plunges, glucose monitors, biological-age clocks) are supported mostly by mechanism and mice. The report's job is to keep those two categories from being confused.

THE THREE EPISTEMIC RULES THAT GOVERN EVERY CLAIM BELOW

(1) Predictor ≠ lever. A biomarker that predicts death is not automatically something that, when changed, prevents it. **(2) Cohort ≠ RCT.** You cannot randomize fitness, sleep, or smoking across decades; healthy-user and reverse-causation bias inflate the strongest-looking numbers. **(3) "Something beats nothing" is the most robust signal in the corpus** — the steepest gains sit at the low end of every dose-response curve; optimization past that is real but smaller and noisier than it is marketed.

Every assertion resolves to a graded entry of the form `claim-id` → `02-domains/*-claims.json` (effect size, population, tier, DOI, confidence note). Disagreements are first-class objects in `06-evidence/CONFLICTS.md`. Tiering follows a ten-rung evidence ladder from `meta` (systematic review of RCTs) down through `rct · cohort · mechanistic · animal · in-vitro · n=1 · anecdotal` to `speculative`. The corpus is the substrate; this document is the read-out.

The corpus in one table

The tier distribution is the headline. Of 247 graded claims, only a minority sit at the rigorous end; the field's loudest interventions cluster in `mechanistic` and `animal`.

EVIDENCE TIER	MEANING	WEIGHT IN CORPUS
<code>meta</code> / <code>rct</code>	Systematic reviews of, and individual, randomized human trials	~56 claims — the rigorous core; concentrated in lipids, exercise, sleep restriction, a few diet trials
<code>cohort</code> / <code>cross-sectional</code> / <code>case-control</code>	Observational human data — associations, not proof	~60 claims — where the <i>strongest-effect</i> levers (fitness, strength, social ties) actually live
<code>mechanistic</code> / <code>animal</code> / <code>in-vitro</code>	Plausible biology; model organisms; cells	~70 claims — where the <i>loudest</i> interventions (NAD ⁺ , senolytics, reprogramming, cold) actually live
<code>n=1</code> / <code>anecdotal</code> / <code>theoretical</code> / <code>speculative</code>	Self-experiment, testimony, untested models	~16 claims — quantified-self protocols and contested fringe biophysics

The asymmetry is the whole story: **strong interventions, weak evidence tier; weak interventions, strong marketing**. Longevity is a field where the best things to do are proven by observation and the most-sold things are proven by mechanism.

I. What the evidence actually supports — the high-confidence levers

The honest ranking is dull on purpose. These eight individually outrank every exotic intervention in the corpus, and they are mutually reinforcing. A program that does not start here is optimizing noise.

- 1 **Don't smoke.** The single largest modifiable mortality factor. A quietly devastating corollary from longevity genetics: the celebrated "longevity genes" of lifespan GWAS are largely *smoking-behavior and cardiometabolic* genes — the heritability of long life is substantially the heritability of not doing the obviously harmful thing. [timmers-2019-parental-lifespan-gwas](#)
- 2 **Build and keep cardiorespiratory fitness (VO₂max).** The strongest exercise–mortality relationship in preventive medicine: roughly **5× lower all-cause mortality** elite-vs-low fitness, ~13% per MET, with **no observed upper limit of benefit** — a magnitude rivaling or exceeding smoking and diabetes. Treat CRF as a clinical vital sign. (Cohort-tier — unrandomizable — but so large and dose-graded it is as close to certain as the field offers.) [crf-vo2max-strongest-mortality-predictor](#)
- 3 **Resistance-train for strength.** It is *strength and grip* — *not muscle mass* — that independently predicts mortality; resistance activity ~10–17% lower mortality, on a **J-curve where more is not better** (benefit peaks ~30–60 min/week). The sarcopenia field itself reframed from mass to strength on exactly this evidence. [grip-strength-mortality-pure](#) [dexa-strength-not-mass-predicts-mortality](#)
- 4 **Move more; break up sitting.** The steepest dose-response is at the sedentary → active transition, and because it is device-measured it is the **least-confounded** signal in the entire domain. [physical-activity-dose-response-mortality](#)
- 5 **Lower lifetime apoB / LDL.** One of the **only causal blood levers** in the corpus — Mendelian genetics, epidemiology and RCTs converge, and because *cumulative* exposure is what matters, earlier is structurally better than later. apoB outperforms LDL-C as the measure. [ldl-apob-causal-ascvd](#) [apob-superior-to-ldlc](#)
- 6 **Sleep ~7 hours, regularly.** U-shaped mortality with a ~7h nadir — both short *and* long sleep worse — with regularity and timing carrying independent weight. The causal direction is partly contested (illness drives both tails). [sleep-duration-mortality-ushape](#)
- 7 **Keep a healthy metabolic profile.** Diabetes ~2× vascular risk; HbA1c predicts within the non-diabetic range; visceral fat predicts beyond BMI. Metabolic health is upstream of most of the others. [hba1c-predicts-cvd-non-diabetic](#) [visceral-fat-independent-mortality-predictor](#)
- 8 **Protect social connection.** Among the largest psychosocial mortality associations in the literature — an effect size that rivals smoking cessation (Holt-Lunstad meta-analyses) — and routinely omitted from optimization stacks *precisely because nothing is sold for it*. Its omission is itself a bias the corpus is built to correct. [social-connection-mortality-meta](#)

THE SHAPE OF THE TRUTH

The high-confidence levers are **functional and behavioral** (fitness, strength, movement, sleep, connection) plus **two causal blood levers** (apoB, glucose). The most useful *measurements* mirror this: they are functional (VO₂max, grip, gait speed, chair-rise, the 10-second balance test) plus a few causal/early blood markers (apoB, Lp(a) measured once, HbA1c, fasting insulin) — not the commercial age-clocks and continuous monitors.

II. Promising but unproven — real signal, lower tier

Worth doing or watching, but honestly tiered down — because the endpoint is a surrogate, the dose studied differs from the dose sold, or the evidence is a single cohort. This is the zone where most defensible "optimization" actually lives.

INTERVENTION	THE REAL SIGNAL	THE HONEST CAVEAT
Zone 2 + VO₂max intervals	Both efficiently raise CRF, the Tier-I lever	"Zone 2 is <i>uniquely</i> optimal for mitochondria" is an over-extrapolation; HIIT drives strong biogenesis too OPEN
Sauna / heat stress	Dose-response to mortality & dementia + plausible HSP/CV mechanism	From one Finnish men's cohort , healthy-user bias unexcluded, no RCT. Infrared saunas do <i>not</i> inherit this evidence
Protein adequacy + leucine threshold	Protects muscle & survival, especially in older adults	A genuine mid-life vs late-life tradeoff with IGF-1/mTOR; not "more is always better" (IGF-1 is U-shaped vs mortality)
Early time-restricted eating	Early-window circadian alignment improves insulin sensitivity & BP	Most TRE benefit is just the caloric restriction it causes ; matched calories erase the edge; late 16:8 is null
Light hygiene (bright AM, dim PM)	Strong melatonin action-spectrum mechanism	Thinner outcome data; blue-blocking <i>glasses</i> specifically show no clear benefit (avoidance ≠ amber lenses)
Modest caloric restriction	CALERIE — the only long-term human CR RCT	~12% CR, surrogate endpoints, moved only one epigenetic clock. Modest and honest
Slow breathing / HRV practice	Real acute autonomic effects; RCT-rich	Surrogate, short, small studies; longevity outcome unproven
Microbiome interventions	Young → old FMT extends lifespan & reverses inflammaging <i>in animals</i>	All human data surrogate; cause-vs-consequence unresolved OPEN
Yoga / meditation	Reliable acute stress/BP/HRV effects	Unblindable, expectancy-prone, allegiance-biased (esp. TM); Goyal 2014 finds only small effects vs <i>active</i> controls

III. Hype & overclaim — the laundering gap

Each item below is sold harder than its evidence. The corpus's single most common failure mode is the **laundering gap**: a real *mechanism*, or a *mouse* result, is marketed as a hard human *outcome* it has not earned. Naming the gap is the antidote.

NAD⁺ boosting (NR / NMN)

NAD⁺ decline with age is real; human RCTs move only surrogates. No demonstrated longevity or hard-endpoint benefit. *Mechanism real, outcome unproven.* [nad-precursor-nr-human-surrogate](#)

Resveratrol / sirtuin activation

The in-vitro SIRT1 activation was substantially a **fluorophore assay artifact**; lifespan extension fails in lean animals. [conflict-resveratrol-sirtuin](#)

Senolytics (D+Q, fisetin)

Striking in mice; human evidence is tiny first-in-human pilots. Not established for healthy people. [senolytics-extend-function-mouse](#)

Metformin / rapamycin off-label

Metformin = a confounded cohort signal plus a trial (TAME) designed but not yet run; rapamycin = mouse lifespan plus one immune RCT, human geroprotective dose unknown. [conflict-rapamycin-dosing](#)

Partial epigenetic reprogramming

Mouse only. Genuinely exciting biology; **not a human intervention.** [partial-reprogramming-ocampo-2016](#)

Cold plunge "for longevity"

Rich BAT/norepinephrine mechanism, but the one human metabolic *outcome* used **prolonged mild cold (hours)**, not the brief plunge sold — a dose↔evidence mismatch. [cold-acclimation-in-sulin-sensitivity-t2d](#)

Wim Hof method

One RCT shows trainable immune suppression — but acute-adrenaline-driven, bundled, small, short, healthy subjects. ⚠ **Lethal in/near water.** [conflict-wim-hof-mechanism](#)

CGM / "glucose spikes" for the healthy

No outcome RCT in non-diabetics; glucose variability has no proven outcome meaning; sensors disagree with each other. [conflict-cgm-healthy-utility](#)

Seed-oil panic

Polarized, low-rigor; higher-tier evidence runs the *other* way (PUFA replacing saturated fat lowers CHD). Certainty exceeds evidence on both sides. [conflict-seed-oils-linoleic-acid](#)

Biological-age clocks as a scorecard

Predict at the population level but are **not a validated surrogate**; first-gen clocks are noisy, clocks disagree, one number can't tell you if a protocol "worked." [conflict-which-clock-is-valid](#)

Consumer HRV / sleep-stage numbers

Wearables measure sleep stages poorly *and* in a flattering direction; overnight HRV is a within-person trend only. "Orthosomnia" is a documented harm.

Blue Zones as proof of a lifestyle

Extreme-age records partly track clerical error and pension fraud; verified demography collapses the rates. The strong claim is contested. [conflict-blue-zones-data-quality](#)

THE 10,000-STEP EXAMPLE, IN ONE LINE

The step target is a 1960s Japanese marketing figure; the mortality benefit **plateaus at ~7,000–8,000 steps**, and intensity adds nothing over volume. A perfect miniature of the whole field — a sold number that the evidence does not support, doing real but smaller good than advertised.

IV. The contested core — 35 first-class conflicts

Disagreement is treated as data, not noise. The corpus holds **35 conflict objects**; roughly **half remain fully OPEN** — usually because no human longevity RCT exists or is feasible to run. None is ever marked "closed"; resolution is recorded, not deleted. The conflicts that would most change the picture if resolved:

CONFLICT	SIDE A	SIDE B	STATUS
Protein ↔ mTOR ↔ longevity	High protein → IGF-1/mTOR → faster aging & cancer	Protein protects muscle/survival, esp. elderly	OPEN likely age-dependent
Do NAD⁺ precursors move a hard endpoint?	Raise NAD ⁺ & improve surrogates → benefit	Only surrogate moved; hype exceeds data	OPEN
Which biological-age clock is valid?	2nd-gen clocks strongly predict mortality	Correlative; clocks disagree; poor reliability	OPEN gates the whole age-test market
mtDNA mutations cause normal aging?	Mutator mice age early; mutations accumulate	Mutator loads far exceed human aging; ROS-independent	OPEN
Microbiome — cause or consequence?	Dysbiosis drives inflammaging; FMT helps (animal)	Composition is downstream of diet/drugs/health	OPEN
Sleep duration — causal or reverse?	Restriction causes real metabolic harm	U-shape: illness/frailty drive both tails	OPEN
Sauna — benefit or healthy-user marker?	Dose-dependent mortality reduction + mechanism	Single male cohort; bias unexcluded; no RCT	OPEN
Metformin geroprotection in non-diabetics?	Users outlive non-diabetics; AMPK/mTOR	Confounded; blunts exercise; no hard-endpoint RCT	OPEN
Free-radical theory of aging	ROS drive aging; antioxidants should help	Mitohormesis: low ROS beneficial; antioxidants null	RESOLVED AGAINST
Resveratrol = direct SIRT1 activator?	CR-mimetic, extends lifespan	In-vitro activation an artifact; no extension in lean animals	OPEN weight against
CR survival in primates	Reduces mortality (Wisconsin)	Healthspan but not survival (NIA)	CONTEXT-DEPENDENT
Seed oils / linoleic acid harmful?	OXLAMs → oxidation/inflammation → CHD	PUFA-for-saturated lowers CHD; bio-marker tracks lower CVD	OPEN weight against toxicity
CGM benefit for the healthy	Response is individual; hidden dysglycemia	Only surrogate; no outcome RCT; excursions benign	OPEN
Blue Zones — real or artifact?	Validated demography found a real cluster	Records track clerical error / pension fraud	OPEN against strong claim

Full register (all 35, both sides, tiers, DOIs, resolution notes): [06-evidence/CONFLICTS.md](#) + [CONFLICTS-REGISTER.md](#). 59 of the graded claims carry a **conflicts_with** edge into these objects.

V. The structural finding — bioenergetics as the bridge to the canon

Above the individual debates sits a structural result that emerged from the research itself. Every actionable longevity domain — exercise (mitochondrial biogenesis, VO_2max as integrated mitochondrial capacity), metabolism (substrate load, ketone signaling), aging (the "mitochondrial dysfunction" hallmark), genetics (mtDNA heteroplasmy), thermal (UCP1 uncoupling) — **reaches up to the same place: the proton-motive force across the mitochondrial membrane.**

That place is not an outcome claim. It is a *law* — Peter Mitchell's chemiosmosis. The corpus's own structure therefore forced a question the canon exists to answer: which findings are **foundations** (axioms, laws, primary derivations) and which are **outcomes** (applications that consume a foundation)? The bioenergetics lineage is the bridge precisely because it sits on the foundation side of that line while every health domain sits on the outcome side and reaches up to it. Critically, it is the *uncontested* spine that connects the inherited — and more speculative — biophysics layer (structured water, biophotons, non-native EMF) to textbook mitochondrial medicine, letting the canon hold a 05-biophysics branch that is rigorous at its core and explicit about its frontier.

ACTED ON THIS REPORT CYCLE — CANON PROMOTIONS

Three foundation-tier figures were promoted into [bucket-canon/05-biophysics](#): **Hans Krebs** (citric-acid cycle), **Jennifer Moyle** (co-proved chemiosmosis; an attribution-parity correction), and **William Martin** (alkaline-vent origin of life, with Lane) — joining Mitchell, Margulis, Lane, Wallace and Szent-Györgyi already carded. A new concept node, **chemiosmosis / proton-motive force / redox bioenergetics**, now anchors the foundation that the outcome layer cites upward to.

The inherited Kruse-tier biophysics is handled by the same discipline rather than by dismissal: the **circadian-light spine is validated** by mainstream chronobiology (melanopsin → SCN; evening blue light disruptive), while its **extensions remain speculative** (sunlight's metabolic primacy over food, UV/IR systemic therapy, deuterium/water coupling). Foundation and frontier, labeled — not merged, not excluded.

VI. The human & institutional landscape

The corpus carded **176 figures** into a 180-node, 155-edge relationship graph spanning mentor lineages (Guarante → Sinclair/Imai; Kenyon → Ruvkun; Mitchell → Lane), institutional affiliations, and documented rivalries. They are indexed without endorsement — mainstream and fringe sit on the same list, separated by an **evidence_posture** tag (*mainstream-rigorous*, *clinical-translator*, *mainstream-communicator*, *frontier-contested*, *fringe-to-canon*, *practitioner-n1*), never by exclusion.

Institutions & the trial reality

24 labs are mapped across academia (Buck, Salk, Harvard, Stanford, UW, Berkeley, Tufts, Einstein, USC), industry (Altos, Calico, Retro, NewLimit, BioAge, Unity, Loyal) and funders (NIA, Hevolution, SENS/LEV, Methuselah). The single most important institutional finding is sobering: across the 15 mapped human aging trials, **every active trial uses a surrogate endpoint** (VO₂max, DXA fat, gait speed, an epigenetic clock). The one trial proposing a hard morbidity/mortality endpoint — **TAME (metformin)** — remains unfunded and unregistered. There is, as of this writing, **no hard-endpoint human longevity trial running anywhere**. This is why so many claims in Section III are structurally stuck at **animal** or **surrogate**: the experiment that would promote them has not been run.

VII. The capstone — read this if nothing else

The evidence is lopsided toward a short list of **boring, powerful, mostly-functional levers**: don't smoke; build and keep cardiorespiratory fitness and strength; move more; sleep ~7 hours regularly; keep apoB/LDL low across life; keep a healthy metabolic profile; protect social connection. These individually outrank every exotic biohack in the corpus, and they compound.

The one causal, modifiable *blood* lever is **apoB/LDL** (measure **Lp(a)** once). The highest-signal *measurements* are **functional** (VO₂max, grip, gait, chair-rise, the 10-second one-leg stand — each an independent mortality predictor) plus a few causal/early blood markers (apoB, Lp(a), HbA1c, fasting insulin). Almost everything that gets *sold* — biological-age clocks, CGM for the healthy, consumer HRV and sleep-stage numbers, senolytics / NAD⁺ / rapamycin for healthy people, cold plunges, seed-oil panic — is either a **correlate dressed as a scorecard**, a **mouse result not yet in humans**, or a **dose that does not match the studied dose**.

The operating instruction: spend attention on the Tier-I levers; treat the Tier-II set as trends worth following; enjoy the Tier-III items as experiments, not protocols. And above the practice, hold the discipline that produced this report — keep foundation-tier *laws* separate from outcome-tier *applications*, grade every claim by what it has actually earned, and let the disagreements stand as objects rather than collapsing them into false certainty.

PROVENANCE & REPRODUCIBILITY

This report is a synthesis layer over `_intake/health-longevity-fitness` (committed to `bucket-foundation`, mirrored to `gdrive:.../research/longevity-fitness-canon/`). Every number is a live count as of 28 June 2026; effect sizes, populations and DOIs live in the linked `02-domains/*-claims.json`; conflicts in `06-evidence/CONFLICTS.md`. The corpus is idempotent — re-running the research waves converges rather than duplicates. Doctrine throughout: **index all · exclude nothing · grade everything · the grade is the neutrality**.

— End of report · Nucleus Brain · AG Farms Venture Studio · bead bkt-bg6 —